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# Effect of the Chemical Structure of a Self-Assembled Monolayer on the Gas-Sensing Behavior of $SnO_2$ Nanowires

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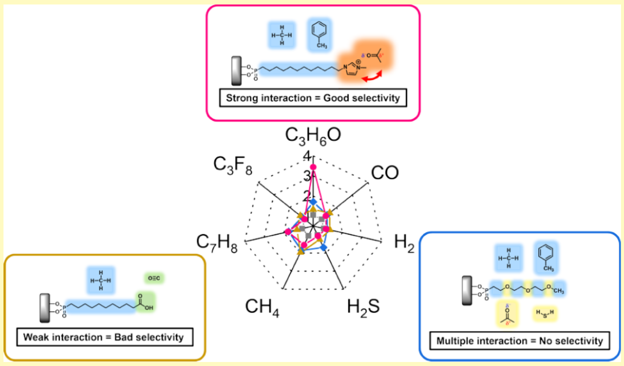
**ABSTRACT:** In this study, detailed investigations of the selective sensing capability of semiconducting metal oxide (SMO)-based gas sensors with self-assembled monolayer (SAM) functionalization were conducted. The selective gas-sensing behavior was improved by employing a simple and straightforward postmodification technique using functional SAM molecules. The chemical structure of the SAM molecules promoted interaction between the gas and SAM molecules, providing a gas selective sensing of $SnO_2$ nanowires (NWs). In addition, a bundle of $SnO_2$ NWs provided a large surface area that could act as a sensing site. SAM functionalization was confirmed by infrared spectroscopy and thermogravimetric analysis, and the selective gas-sensing behaviors were investigated under different sensing conditions. With variations in the chemical structures of the SAM molecules, the selective gas-sensing behaviors also changed as the corresponding intermolecular interaction forces were different. This integration of selective gas sensing and passivation of a sensing surface provides a straightforward approach for the preferred gas sensing of $SnO_2$ NWs. Furthermore, owing to the universal binding characteristics of SAM molecules to the metal oxide surface, this approach can be expanded to other SMO-based gas-sensing platforms.




**KEYWORDS:** *gas sensor, $SnO_2$ nanowire, self-assembled monolayer, selectivity, Surface chemistry*

Technological developments and advancements have improved the quality of life; however, they have concurrently led to concerns regarding the emission of atmospheric pollutants, such as volatile organic compounds (VOCs) and greenhouse gases, which are believed to be the primary causes of global warming and health risks.[1−6] Accordingly, there is a need for suitable monitoring methods to monitor gas pollutant emissions, which has led to the establishment of gas-sensing platforms to detect harmful gases in marginal amounts. In this regard, many studies have reported methods to improve the gas-sensing behavior, either by increasing the sensitivity or selectivity of gas-sensing materials. Among the gas-sensing materials, semiconducting metal oxide (SMO)-based gas-sensing materials have been studied most extensively because they possess a relatively simple gas-sensing mechanism, termed subsequent resistance modulation.[7,8] Other advantages of SMO materials, such as an abundance of material sources, excellent thermal and chemical stability, low toxicity, and good electron mobility, make them promising candidates for gas sensor applications.[9,10]

Furthermore, transforming SMOs into low-dimensional nanostructures, such as zero-dimensional nanoparticles (NPs) or one-dimensional nanowires (NWs), would increase the surface-to-volume ratio, thereby further enhancing the gas-sensing behavior.[11] Owing to these advantages, many different SMO materials have been studied for gas-sensing applications. $SnO_2$ NWs are one of the most promising candidates because they possess a high surface-to-volume ratio, superior physical and chemical stabilities, and good charge carrier mobility, thereby providing good gas-sensing performance.[12−14] However, certain limitations need to be overcome for $SnO_2$ NWs to meet the high standards of state-of-the-art gas sensors. In their pristine form, $SnO_2$ NWs show limited gas-sensing selectivity; they show good capability to distinguish between oxidizing gases and reducing gases, but inferior gas-sensing behavior for more target-specific detection. Therefore, further postmodification processes are required to achieve more target-specific detection with high sensitivities; these processes include noble metal sensitization,[15,16] doping,[8,17,18] ion implantation,[19] and heterojunction formation.[20] In addition, the modulation of oxygen adsorption sites on the surface of $SnO_2$ NWs has been reported to enhance sensor responses.[21] However, all these post-treatment methods require highly complex processes,

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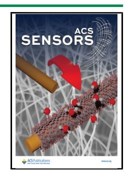





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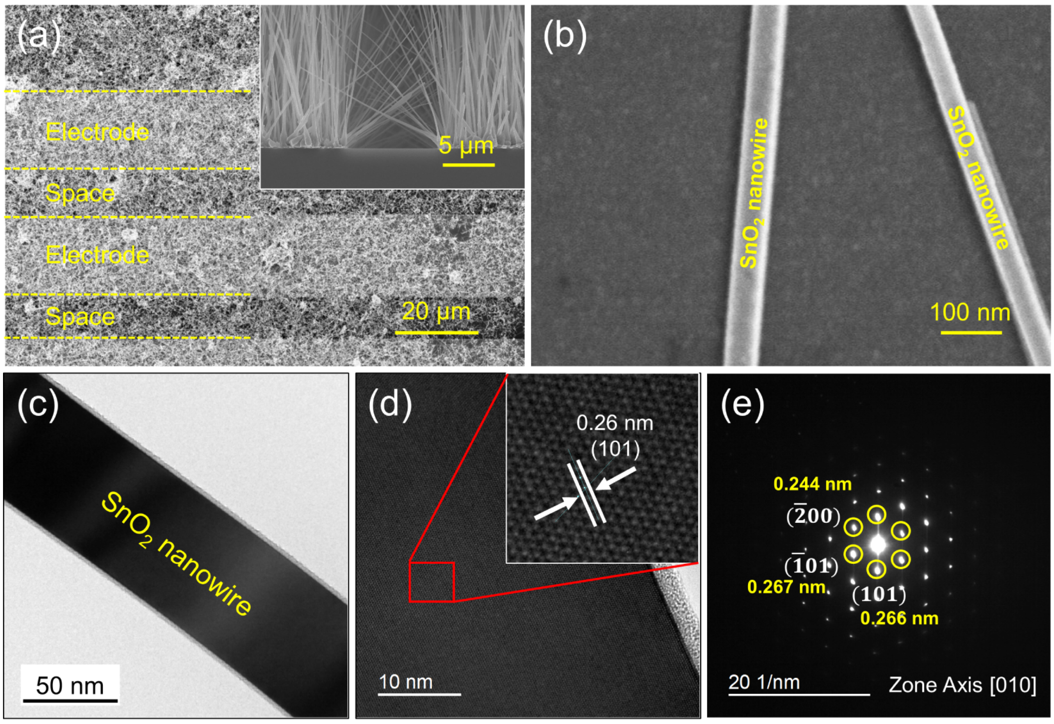


**Figure 1.** (a, b) SEM images of $SnO_2$ NWs used in this study. (c) TEM and (d) HR-TEM images of $SnO_2$ NWs and (e) corresponding SAED pattern.

thereby limiting the use of $SnO_2$ NW-based gas sensors to lab-scale applications.

Owing to the high surface-to-volume ratio of nanomaterials, the contribution of surface properties to the behavior of nanomaterials is substantial. To maintain the properties of the core nanomaterials, the surface properties can be tuned using surface modification methods.[22−24] In this regard, the application of self-assembled monolayers (SAMs) to the surface of nanomaterials could offer substantial advantages via surface modification.[25] As surface modification agents, SAMs have been intensively studied for various applications,[26−29] and have been reported to impart additional functionalities to nanomaterial surfaces. Therefore, when SAM functionalization is applied to SMO surfaces, it is expected that the chemical affinity on the surface can be tuned, thereby promoting chemical-specific interactions on the SAM-modified surfaces and enhancing their selectivity in gas-sensing applications.

Previously, we reported that SAM-modified surfaces showed different gas-sensing behaviors as their functionalities changed.[30] Specifically, when the chemical structures of SAM molecules matched those of the target gas molecules, $SnO_2$ NWs showed superior sensing behaviors. However, further investigation of the relationship between the chemical structures and preferential gas-sensing behaviors is required for a better understanding of the underlying mechanism of the SAM molecule-driven selectivity in the gas-sensing behavior of $SnO_2$ NWs. Therefore, in this study, we investigated the influence of the chemical structure of SAM molecules on the gas-sensing behavior of $SnO_2$ NWs, focusing in detail on the relationships among different chemical moieties and gas molecules. Three types of SAM-functionalized $SnO_2$ NWs were analyzed using different gases. The selective gas-sensing behavior varied as the functionality of the SAM molecules changed. The microstructure of the $SnO_2$ NWs was characterized by scanning electron microscopy (SEM). The SAM functionalization of the $SnO_2$ NWs was confirmed by Brunauer−Emmett−Teller (BET) analysis, Fourier transform infrared (FTIR) spectroscopy, X-ray photoelectron spectroscopy (XPS), and thermogravimetric analysis (TGA).

## EXPERIMENTAL SECTION

**Preparation of $SnO_2$ NW-Based Gas Sensors.** A Au-catalyzed vapor−liquid−solid (VLS) method was used to synthesize $SnO_2$ NWs using previously reported procedures.[15,31] A 200 nm-thick thermally grown $SiO_2$ layer on a Si wafer was used as the substrate, and standard liftoff process was conducted with photolithography and sputtering of Ti (50 nm), Pt (200 nm), and Au (3 nm) in sequence, for creating an electrode pattern. Sn powder (99.9% pure, Sigma-Aldrich) in a ceramic container was placed inside a quartz tube furnace and followed by the heating to 900 °C. The growth of $SnO_2$ NWs on the patterned trilayer electrode was performed for 5 min under a steady flow of $N_2$ (300 sccm) and $O_2$ (10 sccm) gases.

**Functionalization of the SAM on $SnO_2$ NW Surfaces.** The surfaces of the $SnO_2$ NWs were functionalized with 1-methyl-3-(dodecylphosphonic acid)imidazolium bromide (PA-$C_{12}$IMI), 10-carboxydecylphosphonic acid (PA$C_{10}$CA), and (2-{2-[2-methoxy−ethoxy]-ethoxy}-ethyl)phosphonic acid ($PAG_3$) SAM molecules following a previously reported procedure.[32] All the SAM molecules were purchased from SiKÉMIA, France, and were used without further purification. To activate the $SnO_2$ surfaces, $O_2$ plasma (Diener Electronic Pico system) was treated at 200 W, 40 kHz under chamber pressure of 0.3 mbar for 10 min. Subsequently, the $SnO_2$ NWs were immersed in 0.2 mM isopropanol solutions containing the corresponding SAM molecules for 18 h, followed by rinsing with isopropanol and deionized water three times and blow-drying with $N_2$.




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**Characterization.** SEM (FE-SEM, Hitachi S-4200) and TEM (Carl Zeiss LIBRA 200 MC) were used to analyze the microstructure of the $SnO_2$ NWs. For this, sonicated the $SnO_2$ NWs solution in isopropanol was drop-cast onto a Cu TEM grid (HC300-CU, Sigma-Aldrich). SAM functionalization was confirmed by FTIR (IR Prestige 21, Shimadzu) and TGA (TG 209 F1 Libra, Netzsch) analyses. 50 FTIR spectra were obtained with a MIRacle Single Reflection ATR setup featuring a diamond/ZnSe crystal stage and the averaged spectrum was plotted in the 4000−800 $cm^{-1}$ range with a resolution of 4 $cm^{-1}$. In order to determine the amount of SAM molecules bounded on the $SnO_2$ NW surfaces, TGA measurements were obtained from 30° to 900 °C under 40 sccm of $N_2$ and 10 sccm of $O_2$ to resemble an air atmosphere. Further chemical analysis on the chemical states and binding energies were conducted by XPS (KAlpha, Thermo Scientific) with an Al K$\alpha$ X-ray source ($h\nu$ = 1486.6 eV). BET (ASAP 2020, Micromeritics) determined the surface area of the $SnO_2$ NWs.

**Gas-Sensing Measurements.** All gas-sensing experiments were performed with a lab-made customized gas-sensing system, as reported previously.[21] Briefly, the gas chamber was installed in a quartz tube furnace, and the sensing temperature was controlled in the range of 25−350 °C. Mass flow controllers (MFCs) were used to control the gas concentrations by modulating the ratios of dry air and sensing gases. Dry air was purchased from Daeduk Gas Co., Korea, consisting of 21% $O_2$ and 79% $N_2$, with negligible humidity (purity: 99.999%). All sample gases, namely, hydrogen ($H_2$), carbon monoxide (CO), acetone ($C_3H_6O$), octafluoropropane ($C_3F_8$), toluene ($C_7H_8$), methane ($CH_4$), and hydrogen sulfide ($H_2S$), were prepared at different concentrations and mixed with the dry gas and target gases. In all cases, the total flow of the gas mixtures was set to 500 sccm. A Keithley Series 2400 source meter was used to measure the gas sensor resistance under a continuous flow of either air ($R_a$) or the target gas ($R_g$). Using the measured resistance values, the sensor responses were derived as ratios of $R_a$ to $R_g$. For the humidity study, dry and humid air (RH 0−90%, measured at 20 °C) flowed to the sensor chamber at an operating temperature of 175 °C. The temperature and RH were measured using an RH probe before implementing these conditions in the sensor chamber. At 100% RH, dry air (0% RH) flowed into a glass beaker, filled with water, to generate water-saturated air bubbles. These generated bubbles were then mixed with dry air (0% RH) to obtain the desired RH values. The ratios of 100% RH air to dry air for achieving 30%, 60%, and 90% RH values were 3:7, 3:2, and 9:1, respectively. The response time ($\tau_{res}$) was defined as the time taken for the resistance to reach 90% of its final value after the target gas was introduced, while the recovery time ($\tau_{rec}$) was defined as the time taken for the resistance to reach 90% of its initial value after the target gas flow was stopped.

## RESULTS AND DISCUSSION

**Characterization of $SnO_2$ NWs and SAM-Functionalized Surfaces.** Figure 1 shows the SEM and TEM images as well as selected area electron diffraction (SAED) pattern of the $SnO_2$ NW-based gas sensor device. As shown in Figure 1a,b, the length of few tens of micrometer $SnO_2$ NWs with an average diameter of 55 nm are selectively synthesized on the gas sensor electrode, creating a NW junction between the electrodes. In addition, as shown in the TEM images (Figure 1c), smooth and even surfaces with diameters ranging from 20 to 60 nm are observed. The high-resolution (HR) TEM image in Figure 1d reveals the single-crystalline structure of the synthesized $SnO_2$ NWs without any significant defects. From the TEM image, *d*-spacing was measured to be 0.26 nm, corresponding to the (101) plane.[14] Figure 1e shows the SAED pattern obtained from the red box in the HR-TEM image (Figure 1d), which confirms that the synthesized $SnO_2$ NWs have a rutile phase.

To confirm the success of SAM molecules binding on the surface of the $SnO_2$ NWs, FTIR and TGA measurements were conducted (Figure S1). Figure S1a shows the FTIR-ATR results of the SAM-functionalized $SnO_2$ NWs to verify the SAM molecules binding on the $SnO_2$ surfaces. For the pristine $SnO_2$ NWs, no distinguishable organic footprint peaks are observed in the FTIR spectrum. However, organic footprint peaks are clearly observed after SAM functionalization. As all three molecules contain a phosphonic acid (PA) anchoring group, a broad IR absorption band from 1000 to 1200 $cm^{-1}$, representing the overlapping of P−O and P═O stretching vibrations, was observed in all three spectra.[33−35] This overlapping is originated from the multiple binding modes of the PA group on the metal oxide surface, as three different binding reactions—monodentate, bidentate, and tridentate—could exist simultaneously.[36] Furthermore, asymmetric C−H vibrations near 2960 $cm^{-1}$, symmetric C−H vibrations near 2860 and 2930 $cm^{-1}$, and C−H scissoring vibrations near 1460 $cm^{-1}$, were observed in the spectra, supporting the presence of $-CH_2-$ moieties in the SAM molecules. In addition to the common footprint vibrational signals, distinctive vibrations were observed for each molecule. For $PAG_3$, a symmetric C−O vibration at 1254 $cm^{-1}$ from the glycol chain was present, while strong C═O stretching vibrations at 1635 $cm^{-1}$ from the carboxylic acid functional group in $PAC_{10}CA$, C═N stretching vibrations at 1546 $cm^{-1}$, and N−H stretching vibrations at 3145 $cm^{-1}$ from the imidazolium group in $PAC_{12}IMI$ were observed. In addition, XPS measurements confirmed the presence of P 2p signals from the SAM-functionalized $SnO_2$ NWs (Figures S2−S4), while these were not observed for the pristine $SnO_2$ NWs (Figure S5). These results clearly indicate that the surfaces of the $SnO_2$ NWs were successfully functionalized to exhibit different surface chemistries for each SAM molecule.

To confirm the amount of SAM molecules bound to the surface of the $SnO_2$ NWs, TGA measurements were performed. The TGA profile of the pristine $SnO_2$ NWs showed negligible weight loss (∼0.8%), indicating that the surface of the $SnO_2$ NW was an organic-free state. In contrast, the TGA profiles of $PAG_3$, $PAC_{10}CA$, and $PAC_{12}IMI$ started to show a noticeable weight loss from 185 °C, which continued upon heating to 550 °C, and saturated thereafter (Figure S1b). These TGA results of the SAM-functionalized $SnO_2$ NWs indicate that the organic SAM molecules started to decompose from 185 °C and were completely removed above 550 °C. The weight loss value was measured in the temperature range of 100−600 °C, to avoid possible mass contribution from the remaining solvent; the weight losses were 6.38%, 7.12%, and 8.27% for the $PAG_3$-, $PAC_{10}CA$-, and $PAC_{12}IMI$-functionalized $SnO_2$ NWs, respectively. To determine the number of SAM molecules on the $SnO_2$ NW surface, the grafting densities (GDs) were calculated as follows:[37]




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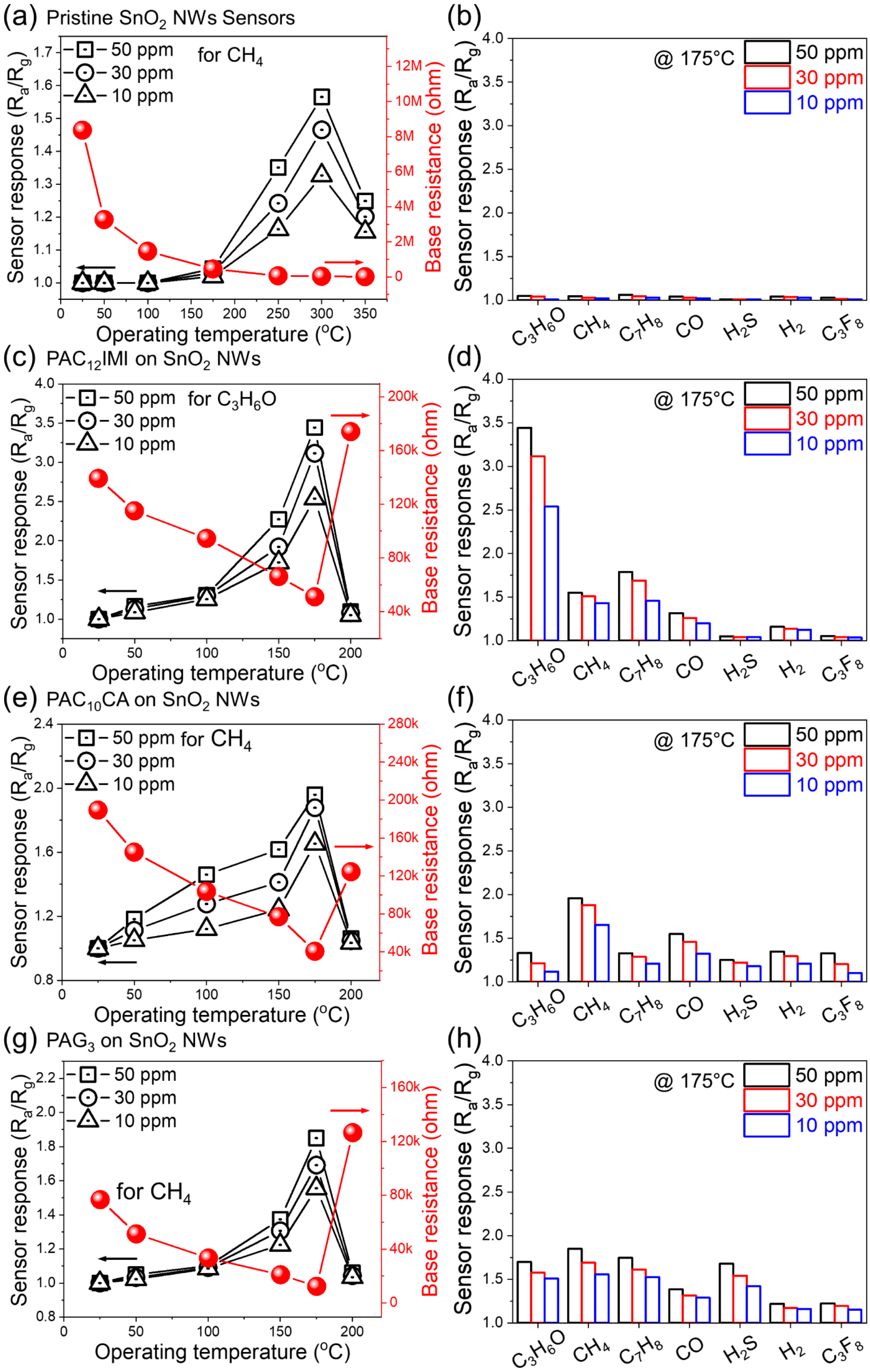


**Figure 2.** Sensor responses and sensing selectivity behaviors of the (a, b) pristine, (c, d) $PAC_{12}IMI$-functionalized, (e, f) $PAC_{10}CA$-functionalized, and (g, h) $PAG_3$-functionalized $SnO_2$ NWs for various concentrations of gas, along with the base resistance as a function of the sensing temperature.

$$GD = \left(\frac{wl}{100 - wl}\right)\left(\frac{6.022 \times 10^{23}}{MW \times SSA}\right)$$

where *wl* represents the percentile weight loss value obtained from the TGA measurements, *MW* represents the molecular weight of the SAM molecules, and *SSA* is the specific surface area obtained from the BET measurements, expressed in $m^2\ g^{-1}$. In this study, the SSA of the $SnO_2$ NWs was determined to be 29.32 $m^2\ g^{-1}$. Using the aforementioned equation, the calculated GD values were 6.13, 5.91, and 4.5 molecules $nm^{-2}$ for $PAG_3$, $PAC_{10}CA$, and $PAC_{12}IMI$ on the $SnO_2$ NW surfaces, respectively.

**SAM-Induced Gas Selectivity of $SnO_2$ NW-Based Gas Sensors.** To determine the optimal sensing temperature of the $SnO_2$ NW-based gas sensor, the sensing performance of pristine $SnO_2$ NWs for $CH_4$ was evaluated. Figure S6 shows the dynamic resistance curves of the pristine $SnO_2$ NW-based gas sensors in 10, 30, and 50 ppm of $CH_4$ gas at various sensing temperatures, ranging from room temperature (RT; 25 °C) to 350 °C. In all cases, the resistance decreased when $CH_4$ gas was injected into the gas chamber. $CH_4$ is known to be a reducing gas, which donates electrons upon adsorption to the surface.[38] The reduced resistance of the $SnO_2$ NWs upon exposure to $CH_4$ gas revealed the *n*-type nature of the $SnO_2$ NWs. Moreover, up to 100 °C, there was no effective response observed upon gas exposure. To gain better insight, the baseline resistance is depicted and the temperature-dependent response of $CH_4$ exposure at different concentrations is plotted in Figure 2a. The baseline resistance decreases as the sensing temperature increases, demonstrating the semiconducting nature of the sensors. An increase in the sensing temperature led to more electrons jumping to the conduction band of $SnO_2$, which in turn caused the overall resistance of the sensor to decrease. Furthermore, the sensing response toward $CH_4$ gradually increased as the sensing temperature increased up to 300 °C, and thereafter, the sensing response started to decrease. Under the low-temperature regime, i.e., below 175 °C, the sensor response was low due to the insufficient energy of the gas to overcome the adsorption barrier. However, at higher temperatures, the adsorption and desorption rates both increase. Moreover, at the optimal sensing temperature, i.e., 300 °C, the maximum net adsorption is observed. With a further increase in the temperature, i.e., above 300 °C, the desorption rate increases above the adsorption rate, and thus, the sensor response decreases.

Similarly, sensing performances were measured for all three SAM-functionalized $SnO_2$ NW sensors, i.e., $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized sensors, to define the optimal sensing temperatures, ranging from RT to 200 °C. Figure S7a shows the dynamic resistance curves of $PAC_{12}IMI$ on the $SnO_2$ NW gas sensor exposed to 10, 30, and 50 ppm of $C_3H_6O$ gas at various temperatures. The resistance decreases upon exposure to gas molecules, indicating that the *n*-type semiconductor behavior of the $SnO_2$ NWs is not affected by SAM functionalization. Figure 2c shows the baseline resistances and sensor responses to different concentrations of $C_3H_6O$ gas as a function of the sensing temperature. Compared to the pristine $SnO_2$ NWs without SAM functionalization, the baseline resistance of the $PAC_{12}IMI$-functionalized sensor exhibits orders of magnitude lower values at RT (pristine: 8.37 MΩ, $PAC_{12}IMI$-functionalized: 0.14 MΩ). This indicates that SAM functionalization successfully passivates the $SnO_2$ NW surfaces, thereby restraining oxygen adsorption on the $SnO_2$ NW surfaces. Therefore, the baseline resistance was lower than that of the pristine $SnO_2$ NW sensor. Moreover, the baseline resistance of the $PAC_{12}IMI$-functionalized sensor decreased as the temperature increased up to 175 °C. A further increase in the temperature led to an increase in the baseline resistance. This increase in the baseline resistance could also be considered evidence of the presence of SAM molecules on the $SnO_2$ NW surface, as the SAM molecules started to decompose after 185 °C (shown in TGA profiles in Figure S1b). Similar to the $PAC_{12}IMI$-functionalized sensor, the $PAC_{10}CA$- and $PAG_3$-functionalized sensors showed the same tendency for the baseline resistance and dynamic resistance curves. Figures S8a and S9a show the dynamic resistance curves of the $PAC_{10}CA$-functionalized sensor and $PAG_3$-functionalized sensor, respectively. Both sensors were exposed to 10, 30, and 50 ppm of $CH_4$ gas at various temperatures from RT to 200 °C. As shown in Figures S8a and S9a, both sensors exhibit *n*-type behavior along with good recovery of the sensing signal. Figure 2e,g shows the baseline resistances and sensor responses of the $PAC_{10}CA$- and $PAG_3$-functionalized sensors to different gas concentrations of $CH_4$ as a function of the sensing temperature. Similar to the $PAC_{12}IMI$-functionalized sensor, the baseline resistances of the $PAC_{10}CA$- and $PAG_3$-functionalized sensors are 0.19 MΩ and 0.08 MΩ, respectively. These results further support the aforementioned mechanism of passivation of the $SnO_2$ NW surfaces. Moreover, the baseline resistances for both sensors gradually decreased upon heating the sensors to 175 °C, and further temperature increases led to increased baseline resistance, indicating the presence of SAM molecules on the $SnO_2$ NW surfaces. As revealed from the TGA profiles in Figure S1b, the removal of the SAM molecules from the surfaces commences at temperatures above 185 °C. As the main aim of this study was the investigation of the effect of the chemical structures on the selective gas-sensing behavior of SAM-functionalized $SnO_2$ NW surfaces, SAM-functionalized $SnO_2$ NWs were not measured at temperatures above 200 °C. Hence, a fixed sensing temperature of 175 °C was used for further selective sensing studies to better compare the chemical structural influence on the gas-sensing behavior of the $SnO_2$ NWs.

To determine the selective gas-sensing behavior of the pristine $SnO_2$ NWs, the sensor was exposed to various gases, and the corresponding response behaviors are plotted in Figures 2b and S6b. Similar to the response to $CH_4$ gas, the sensor responses toward different gases, such as CO, $C_3H_6O$, $H_2$, $H_2S$, $C_3F_8$, and $C_7H_8$, exhibit similar values, indicating that the pristine $SnO_2$ NWs do not possess gas-selective responses. To study the sensor selectivity upon different SAM functionalization, the $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized sensors were exposed to various gases at 175 °C (Figures S7b, S8b, and S9b), and their corresponding selectivity histograms are depicted in Figure 2d,f, and h, respectively. The $PAC_{12}IMI$-functionalized sensor shows a higher response to $C_3H_6O$ gas than to other gas molecules, and the sensor responses to 10, 30, and 50 ppm of $C_3H_6O$ gas are 2.54, 3.12, and 3.44, respectively. The sensor responses toward 50 ppm of different gases, such as CO, $H_2$, $H_2S$, $CH_4$, $C_3F_8$, and $C_7H_8$, decrease to 1.31, 1.16, 1.05, 1.55, 1.05, and 1.79, respectively. SAM-induced selective gas-sensing behavior was also observed for the $PAC_{10}CA$-functionalized sensor, with a higher response to $CH_4$ than to other gases (Figure 2f). However, this preferred response behavior was less pronounced than that of the $PAC_{12}IMI$-functionalized sensor. In particular, $PAC_{10}CA$-functionalized sensor's responses to 10, 30, and 50 ppm of $CH_4$ were 1.65, 1.88, and 1.96, respectively, whereas the sensor responses toward 50 ppm of other gases, such as CO, $H_2$, $H_2S$, $C_3H_6O$, $C_3F_8$, and $C_7H_8$, decreased to 1.55, 1.35, 1.25, 1.33, 1.32, and 1.33, respectively. Although the $PAC_{10}CA$-functionalized sensor showed preferred sensing behavior for $CH_4$ gas, the selective response was not as pronounced as that for the other gas molecules, and therefore, the selectivity was decreased. For the $PAG_3$-functionalized sensor, the gas-selective response decreased further. As shown in its dynamic resistance curves at 175 °C (Figure S9b) and




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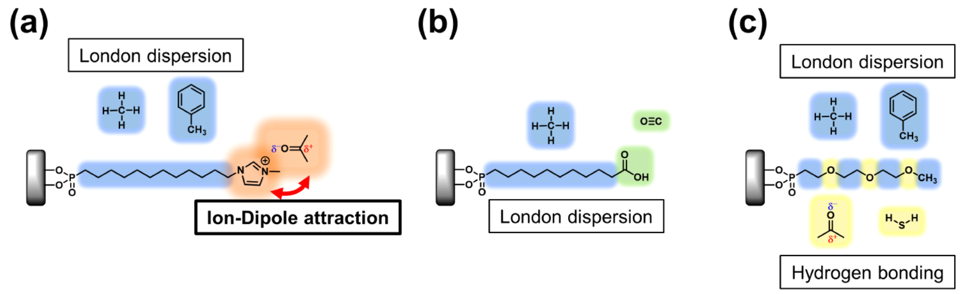


**Figure 3.** Schematics of the selective sensing mechanisms of (a) $PAC_{12}IMI$-, (b) $PAC_{10}CA$-, and (c) $PAG_3$-induced intermolecular interactions.

corresponding selectivity pattern (Figure 2h), the sensor has an improved response toward $CH_4$ gas compared to the pristine $SnO_2$ NW sensor, with values of 1.56, 1.69, and 1.85 for 10, 30, and 50 ppm of $CH_4$ gas, respectively. However, the $PAG_3$-functionalized sensor did not show selective gas-sensing behavior, as it exhibited a similar sensor response upon exposure to other gases; for 50 ppm of CO, $H_2$, $H_2S$, $C_3H_6O$, $C_3F_8$, and $C_7H_8$ gases, the sensor responses were 1.39, 1.22, 1.68, 1.70, 1.22, and 1.75, respectively. The differences between the sensor responses of the most favorable gas ($CH_4$) and those to the other gases was less than 0.7. In particular, the difference was less than 0.2 for $H_2S$, $C_7H_8$, and $C_3H_6O$ gases. Although the overall sensor responses were improved upon $PAG_3$ functionalization at 175 °C, preferential sensing behaviors were not imparted to the $SnO_2$ NWs. To gain a general insight, the SAM-induced gas selectivity patterns of all tested gas sensors are depicted in Figure 4a. The $PAC_{12}IMI$-functionalized $SnO_2$ NW sensor showed the best selective gas-sensing behavior, exhibiting the highest response to $C_3H_6O$ gas. The $PAC_{10}CA$-functionalized sensor showed less-pronounced gas selectivity, whereas the $PAG_3$-functionalized sensor exhibited marginal gas selectivity. This clearly indicates that the preferential interaction between SAM molecules and gas molecules relies heavily on the chemical structure of the functionalized SAM molecules.

**Effect of Chemical Functionality on Gas-Sensing Behaviors.** As discussed in the previous section, the SAM-induced gas selectivity differed among molecules. Specifically, the selective gas-sensing behaviors relied on the chemical structures and functionalities of the SAM molecules. Figure 3 shows a schematic of the basic mechanism of SAM-induced gas selectivity. In Figure 3a, for the $PAC_{12}IMI$ molecule, which contains a positively charged imidazolium group, ion−dipole attraction dominates the intermolecular interaction toward external gas molecules. Therefore, the $PAC_{12}IMI$-functionalized sensor showed superior selective gas sensing toward $C_3H_6O$, which is a polar gas molecule with a dipole moment of 2.69 D. In addition, the $PAC_{12}IMI$-functionalized sensor showed a slightly improved sensing response toward $CH_4$ and $C_7H_8$ gases than other gases: 1.55 for 50 ppm of $CH_4$ and 1.79 for 50 ppm of $C_7H_8$. This is mainly due to the presence of hydrocarbon structures from the dodecyl chain, which contribute to a weak London dispersion attraction force for interaction with the external gas molecules.

Similarly, $PAC_{10}CA$ also contains hydrocarbon structures from the decyl chain (Figure 3b), which promote better interaction with the $CH_4$ gas via a London dispersion attraction force; the corresponding sensor showed a sensor response of 1.96 for 50 ppm of $CH_4$ gas. Meanwhile, the presence of a carboxyl group from $PAC_{10}CA$ also promoted interaction with CO gas molecules with carbonyl affinity matching, and thus, the $PAC_{10}CA$ sensor showed a sensor response of 1.55 toward 50 ppm of CO gas. Nevertheless, the overall selective sensing behavior was diminished because the selective sensing was governed only by a weak London dispersion attraction force for $PAC_{10}CA$, whereas a strong ion−dipole interaction dominated the attraction toward $C_3H_6O$ gas for the $PAC_{12}IMI$ sensor.

However, this intermolecular attraction does not guarantee a preferred interaction toward a selective gas molecule. As shown in Figure 3c, the presence of an ether chain (C−O−C) from $PAG_3$ induces multiple hydrogen bonds toward different gas molecules, such as $H_2S$ and $C_3H_6O$. In addition, the $PAG_3$ molecule contains a methyl group ($-CH_3$) and methylene group ($-CH_2-$), which also trigger attraction toward the $CH_4$ and $C_7H_8$ gas molecules. As a result, the $PAG_3$-functionalized sensor showed improved sensor responses toward $CH_4$, $C_7H_8$, $C_3H_6O$, and $H_2S$ gas molecules, but did not show any prominent gas selectivity.

SAM-functionalization may facilitate the preferred intermolecular interactions near the sensing surfaces. However, SAM molecules were not directly reacted to the corresponding gas molecules. Figures S2−S5 shows XPS spectra before and after the sensing experiments for each $PAC_{12}IMI$-, $PAC_{10}CA$-, $PAG_3$-functionalized, and the pristine $SnO_2$ NWs gas sensors, respectively. In all cases, XPS peaks were slightly shifted to the higher binding energy levels, which occurred frequently as the surface became oxidized.[39] However, SAM molecule induced XPS peaks, i.e., P, C, and N, still observed after the repeated sensing experiments, suggesting SAM molecules were not modified during the gas sensing experiments.

These findings suggest a basic mechanism for the preferred interaction between SAM molecules and specific gas molecules. Among the energies for different intermolecular interactions, the ion−dipole interaction energies typically lie within the range of 50−200 kJ $mol^{-1}$, whereas the hydrogen bond energy varies from 5 to 100 kJ $mol^{-1}$, and the London dispersion force lies within the range of 1−30 kJ $mol^{-1}$. Based on the sensing results reported in the previous section, it can be concluded that a stronger interaction force results in a greater sensor response. Moreover, it is important to choose appropriate chemical structures to achieve the best gas selectivity. Correspondingly, a precise molecular design is required to successfully modulate the SAM-induced gas selectivity.

**Reliability of SAM-Functionalized $SnO_2$ NW-Based Gas Sensors.** To verify the reliability of the SAM-functionalized sensors, the effects of humidity were first investigated, because exposure to moisture is inevitable in practical applications. Figure S10 presents the dynamic resistance

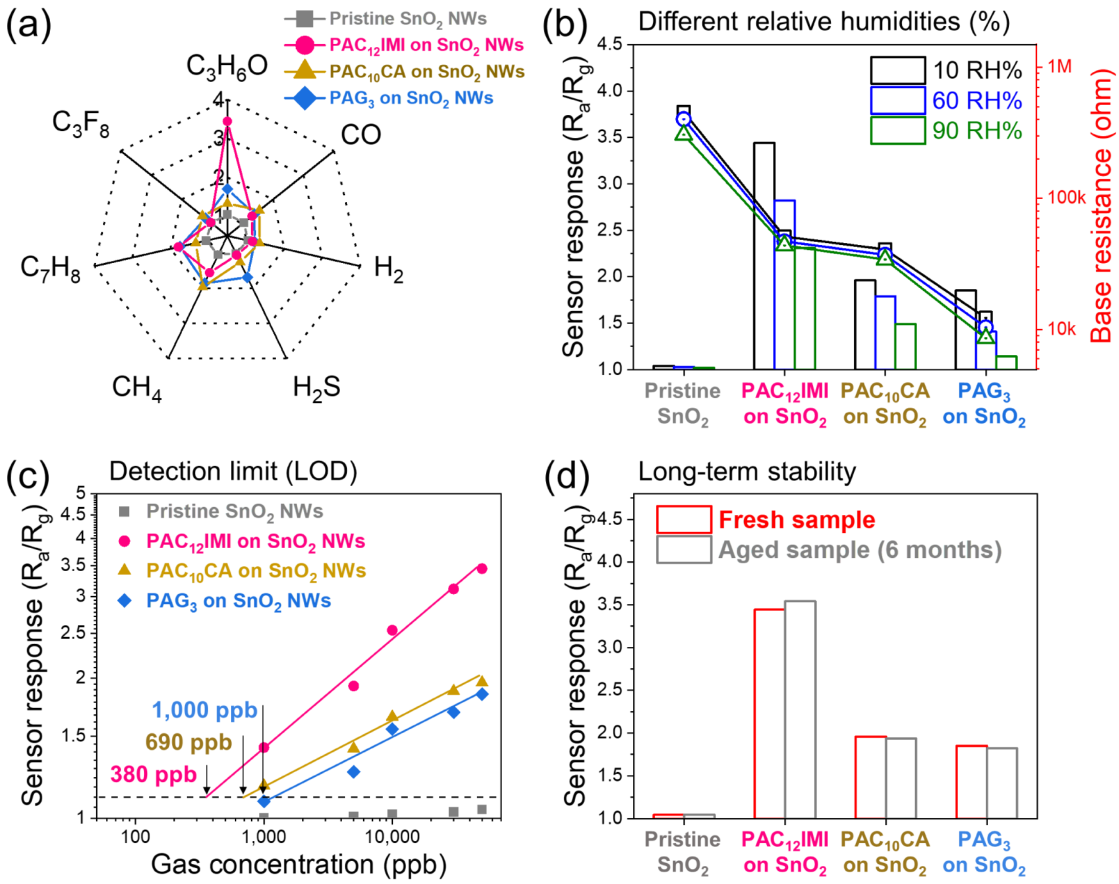


**Figure 4.** (a) Radar chart of different gas sensors to compare the selective sensing responses. (b) Humidity dependence of the sensor responses, (c) detection limits, and (d) long-term stabilities of $SnO_2$ NW gas sensors used in this study.

curves of the pristine, $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized $SnO_2$ NW gas sensors under 10 ppm of the indicated gases at 175 °C with different RH levels (10, 60, and 90%). All the SAM-functionalized sensors were tested for three consecutive sensing measurements. In all cases, the sensor response decreases with increasing humidity. As shown in Figure 4b, the baseline resistance and sensor response decrease as the humidity increases from 10 to 90%. In the presence of humidity, water molecules are inevitably adsorbed on the sensor surface, which compete with the sensing gas molecules as they occupy the adsorption sites. This leads to a decrease in the adsorption of gas molecules onto the sensor surface; therefore, the sensor response decreases. This explanation also applies to the SAM-functionalized sensors. Although the SAM-functionalized layer on the $SnO_2$ NW surface acts as a passivation layer, the SAM molecules do not completely cover the surface of the $SnO_2$ NWs. If every surface was covered with SAM molecules, there would be no effective adsorption sites available on the $SnO_2$ NWs. Nevertheless, all the SAM-functionalized sensors exhibited gas-sensing behavior, indicating that there were still free adsorption sites available on the $SnO_2$ NW surfaces even after SAM functionalization. The humidity dependence of the SAM-functionalized sensors followed a similar decreasing trend as the RH increased. This result illustrates that the principle applicable to the pristine $SnO_2$ NW sensor is also applicable to the SAM-functionalized sensors, whereby the water molecules only affect the adsorption sites of SAM-free $SnO_2$ surfaces. Moreover, the decrease in resistance was due to the supply of electrons from the adsorbed oxygen as they reacted with the water molecules. This relationship can be described by the following equation:[40]

$$H_2O + 2Sn(lat) + O^- \rightarrow 2(Sn - OH) + e^-$$

The limit of detection (LOD) is an important indicator that reflects the sensing performance. The lower LOD is defined as the minimum concentration at which the response differs significantly from the noise signal. Sensor noise can be calculated by changing the relative response of the sensor over the baseline. To determine LOD of the gas sensors, concentration-dependent sensing measurements were conducted for all the gas sensors. Before exposure to gases, take the average value of 20 consecutive data points, and calculate the root-mean-square deviation according to the following equation as 0.00342.[41]

$$RMS_{noise} = \sqrt{\frac{S^2}{N}}$$

Where $RMS_{noise}$ is the root-mean-square error and N is the number of data points. The LOD is defined as the following equation:

$$LOD = 3\frac{RMS_{noise}}{Slope}$$

Figure S11 shows the dynamic resistance curves of the pristine, $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized $SnO_2$ NW sensors for different concentrations (1−50 ppm) of target gases at 175 °C; the corresponding calibration curves




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are plotted in Figure 4c. The LODs values were determined by the extrapolation of the calibration line on the *y*-axis. They were 380, 690, and 990 ppb for the $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized sensors, respectively. According to the definition of the LOD, the theoretically calculated LOD value of the $PAC_{12}IMI$-functionalized sensor was 366 ppb, which was 14 ppb differ from the LOD based on extrapolation on the *y*-axis (Figure 4c). Similarly, the theoretically calculated LOD values for $PAC_{10}CA$- and $PAG_3$-functionalized sensors were 678 and 1032 ppb, respectively, indicating all the measurements were matched well to the calculations.

Similar to the selective sensing behavior that changed upon modulation of the SAM molecules, the LOD values decreased as the SAM molecules showed less selectivity. This is also attributed to the intermolecular interactions facilitating the attraction of gas molecules, as affinity-matching chemical structures are present on the sensor surface. Because the SAM-functionalized surface provides a less-selective chemistry, it hampers the sensor response to gas molecules. Nevertheless, all SAM-functionalized sensors, including the least selective $PAG_3$-functionalized sensor, showed sub-ppm LOD values, further supporting the potential of these gas sensors to detect target gases in real applications.

In addition, the long-term stabilities of the SAM-functionalized sensors were investigated. Figure S12 shows the dynamic resistance curves of the pristine, $PAC_{12}IMI$-, $PAC_{10}CA$-, and $PAG_3$-functionalized sensors under 10 ppm of their corresponding gases at 175 °C in their as-fabricated fresh state and after aging for six months. In order to check the reproducibility, all sensors were tested in three repeating cycles. Regardless of the aging time, the sensing curves remained similar to their fresh states, and as shown in Figure 4d, there are only marginal differences between the responses after six months. This directly reflects the high long-term stability of the gas sensors. The sample-to-sample reliability of the SAM-functionalized $SnO_2$ NWs gas sensors were also studied. Three different gas sensor devices were prepared with same SAM-functionalization conditions, and each of them were measured under identical sensing parameters. Figure S13 shows the representative sample-to-sample reliability behavior of the $PAC_{12}IMI$-functionalized $SnO_2$ NWs sensors. Figure S13a clearly represents that with $PAC_{12}IMI$, overall sensor responses were not differ with the devices. Also, as shown in Figure S13b, the averaged sensor response value of $PAC_{12}IMI$-functionalized $SnO_2$ sensor to the $C_3H_6O$ gas under 175 °C was 3.55 with standard deviation of 0.095. In addition, all three devices exhibited similar base resistance levels, 51.2 k, 52.3 k, and 50.7 kΩ, indicating the stability of the SAM-functionalization.

Table S1 compares gas sensing properties of present sensors with those of other sensors in the literature. Overall, present sensors show comparable or even higher sensing performance relative to the listed sensors in terms of higher response and lower sensing temperature.

## CONCLUSION

In this study, we investigated the influence of the chemical structure of the SAM on the gas-sensing behavior of $SnO_2$ NW-based gas sensor devices. The experimental results demonstrated the tunability of the gas selectivity with different chemical moieties of the SAM molecules. Specifically, intermolecular interactions between the SAM molecules and external gas molecules strongly influenced the preferred sensing behavior. Ionically charged moieties showed a preferred interaction with $C_3H_6O$ gas, whereas alkyl chains favored $CH_4$ and $C_7H_8$ gases. Moreover, the glycol group could interact better with $H_2S$ as well as $C_3H_6O$ as it facilitated hydrogen bonding. Because the energy of the ion−dipole attraction was greater than those of the hydrogen-bonding and London dispersion attraction forces, $PAC_{12}IMI$ showed the best selective gas-sensing behavior toward $C_3H_6O$ gas. $PAG_3$ exhibited the lowest gas selectivity because its glycol chain contained multiple interaction motifs. Finally, reliability studies demonstrated the potential use of the SAM-treated gas sensor devices in real-world applications. These results successfully provide a deeper understanding of the preferred gas-sensing behavior of organic molecules present on the surfaces of the sensing materials. Specifically, we found that different chemical moieties exhibited different interaction behaviors with the gas molecules. We strongly believe that these findings can serve as a foundation for designing highly complicated chemical surfaces to achieve higher sensing selectivities for complex gas molecules. Furthermore, the universal binding characteristics of PAs toward metal oxide surfaces will allow the SAM functionalization strategy to be applied to other metal oxide-based gas-sensing devices.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssensors.4c02105.

Basic characterization of the SAM layer on $SnO_2$ NWs (FTIR-ATR, TGA, and XPS) and dynamic resistance curves for all $SnO_2$ NW-based gas sensor devices (PDF)

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### Author Contributions

H.P.: Conceptualization, Data curation, and Writing−original draft, review and editing, J.-H.K.: Conceptualization, Data curation, and Writing−original draft, review and editing.

### Notes

The authors declare no competing financial interest.

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